

Komputasi Numerik

PERTEMUAN 10



Interpolasi Newton–Gregory, Stirling dan Bessel

2026





Komnum Week 9

Apa Yang Akan Kita Pelajari?

01 Interpolasi Newton-Gregory

02 Interpolasi Stirling

02 Interpolasi Bessel



INTERPOLASI



Jika pada materi pencocokan kurva sebelumnya anda diminta menaksir bentuk fungsi melalui sederetan data, maka sekarang kita diminta untuk **mengestimasi nilai fungsi $f(x)$** di antara beberapa nilai fungsi yang diketahui (tanpa mengetahui bentuk fungsi yang menghasilkannya)

Untuk menaksir harga tengahan diantara titik-titik data yang telah ada :

Polinomial Newton

Polinomial Lagrange

Interpolasi Newton-Gregory

Interpolasi Stirling

Interpolasi Bessel

Tabel Beda Interpolasi Newton-Gregory

s	x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$	$\dots$
0	x_0	f_0					
			Δf_0				
1	x_1	f_1		$\Delta^2 f_0$			
			Δf_1		$\Delta^3 f_0$		
2	x_2	f_2		$\Delta^2 f_1$		$\Delta^4 f_0$	$\dots$
			Δf_2		$\Delta^3 f_1$		
$\vdots$	$\vdots$	$\vdots$					
$\vdots$	$\vdots$	$\vdots$					
$\vdots$	$\vdots$	$\vdots$					
			Δf_{n-3}		$\Delta^3 f_{n-4}$		
$n - 2$	x_{n-2}	f_{n-2}		$\Delta^2 f_{n-3}$		$\Delta^4 f_{n-4}$	$\dots$
			Δf_{n-2}		$\Delta^3 f_{n-3}$		
$n - 1$	x_{n-1}	f_{n-1}		$\Delta^2 f_{n-2}$			
			Δf_{n-1}				
n	x_n	f_n					

Interpolasi Newton-Gregory

Interpolasi Newton-Gregory **Forward** dari atas ke bawah:

$$f(x_s) = f_0 + s\Delta f_0 + \frac{s(s-1)}{2!}\Delta^2 f_0 + \frac{s(s-1)(s-2)}{3!}\Delta^3 f_0 + \dots + \frac{s(s-1)(s-2)\cdots(s-n+1)}{n!}\Delta^n f_0$$

Interpolasi Newton-Gregory :**Backward**: dari bawah ke atas

$$f(x_s) = f_0 + s\Delta f_{-1} + \frac{s(s+1)}{2!}\Delta^2 f_{-2} + \frac{s(s+1)(s+2)}{3!}\Delta^3 f_{-3} + \dots + \frac{s(s+1)(s+2)\cdots(s+n-1)}{n!}\Delta^n f_{-n}$$

$$s = \frac{x_s - x_0}{h}, \quad h = \Delta x$$

Contoh Interpolasi Newton-Gregory

Contoh interpolasi dengan **orde-3**:

Interpolasi Newton-Gregory **Forward**:

$$f_3(x_s) = f_0 + s\Delta f_0 + \frac{s(s-1)}{2!}\Delta^2 f_0 + \frac{s(s-1)(s-2)}{3!}\Delta^3 f_0$$

Interpolasi Newton-Gregory **Backward**:

$$f_3(x_s) = f_0 + s\Delta f_0 + \frac{s(s+1)}{2!}\Delta^2 f_{-1} + \frac{s(s+1)(s+2)}{3!}\Delta^3 f_{-2}$$

$$s = \frac{x_s - x_0}{h}, \quad h = \Delta x$$

➤ Interpolasi Newton-Gregory ➤

Contoh Soal 1

Carilah nilai $f(x_s)$ untuk $x_s = 3$, jika diketahui fungsi tersebut menghasilkan nilai tabel sebagai berikut:

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
2	31			
		351		
5	382		810	
		1161		486
8	1543		1296	
		2457		
11	4000			

Gunakan Interpolasi Newton-Gregory Forward!

➤ Jawaban Contoh 1 ⚡

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
2	31	351		
5	382	1161	810	
8	1543	2457	1296	486
11	4000			

$$f_3(x_s) = f_0 + s\Delta f_0 + \frac{s(s-1)}{2!}\Delta^2 f_0 + \frac{s(s-1)(s-2)}{3!}\Delta^3 f_0$$

$$s = \frac{x_s - x_0}{h}, \quad h = \Delta x \rightarrow h = 3$$

$$s = \frac{3-2}{3} = \frac{1}{3}$$

$$f_3(3) = 31 + \frac{1}{3}(351) + \frac{(\frac{1}{3})(\frac{1}{3}-1)}{2!}(810) + \frac{(\frac{1}{3})(\frac{1}{3}-1)(\frac{1}{3}-2)}{3!}(486)$$

-90

117

30

$$f_3(3) = 88$$

Interpolasi Newton-Gregory

Contoh Soal 2

Diketahui Tabel sebagai berikut:

x	y	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
2	18.00			
		108.53		
2.4	126.53		192.82	
		301.34		196.42
2.8	427.87		389.24	
		690.58		316.11
3.2	1118.45		705.35	
		1395.93		476.34
3.6	2514.38		1181.69	
		2577.62		683.03
4	5092.00		1864.72	
		4442.34		
4.4	9534.34			

- Carilah nilai $f(x)$, ketika $x = 3,3$ dengan menggunakan **Newton-Gregory Forward**
- Carilah nilai $f(x)$, ketika $x = 3,3$ dengan menggunakan **Newton-Gregory Backward**
- Note:
 - Hitung Error jika diketahui $f(3,3) = 1385,77$

➤ Jawaban Contoh 2 ➤

- Newton-Gregory Forward:

x	y	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
2	18.00	108.53		
2.4	126.53	301.34	192.82	
2.8	427.87	690.58	389.24	196.42
3.2	1118.45	1395.93	705.35	316.11
3.6	2514.38	2577.62	1181.69	476.34
4	5092.00	4442.34	1864.72	683.03
4.4	9534.34			

$$f(x_s) = f_0 + s\Delta f_0 + \frac{s(s-1)}{2!}\Delta^2 f_0 + \frac{s(s-1)(s-2)}{3!}\Delta^3 f_0 + \dots + \frac{s(s-1)(s-2)\cdots(s-n+1)}{n!}\Delta^n f_0$$

$$s = \frac{x_s - x_0}{h}, \quad h = \Delta x$$

$$s = \frac{3.3 - 3.2}{h}, \quad h = 3.2 - 2.8 = 0.4$$

$$= 0.25$$

$$f_3(3.3) = 1118.45 + (0.25)(1395.93) + \frac{(0.25)(0.25-1)}{2!}(1181.69) + \frac{(0.25)(0.25-1)(0.25-2)}{3!}(683.03)$$

348.98
-110.78
37.35

$$= 1394$$

➤ Jawaban Contoh 2 ➤

- Newton-Gregory **backward**:

x	y	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
2	18.00	108.53		
2.4	126.53	301.34	192.82	
2.8	427.87	690.58	389.24	196.42
3.2	1118.45	1395.93	705.35	316.11
3.6	2514.38	2577.62	1181.69	476.34
4	5092.00	4442.34	1864.72	683.03
4.4	9534.34			

$$f(x_s) = f_0 + s\Delta f_{-1} + \frac{s(s+1)}{2!}\Delta^2 f_{-2} + \frac{s(s+1)(s+2)}{3!}\Delta^3 f_{-3} + \dots + \frac{s(s+1)(s+2)\cdots(s+n-1)}{n!}\Delta^n f_{-n}$$

$$s = \frac{x_s - x_0}{h}, \quad h = \Delta x$$

$$s = \frac{3.3 - 3.2}{h}, \quad h = 3.2 - 2.8 = 0.4$$

$$= 0.25$$

$$f_3(3.3) = 1118.45 + (0.25)(690.58) + \frac{(0.25)(0.25+1)}{2!}(389.24) + \frac{(0.25)(0.25+1)(0.25+2)}{3!}(196.43)$$

$$= 1374.93$$

Interpolasi Stirling

Interpolasi **Stirling**:

$$f_n(x) = f_0 + s\Delta f + \frac{s^2}{2!}\Delta^2 f_{-1} + \frac{s(s^2-1)}{3!}\Delta^3 f + \frac{s^2(s^2-1)}{4!}\Delta^4 f_{-2} + \frac{s(s^2-1)(s^2-4)}{5!}\Delta^5 f + \dots$$

Diagram annotations for the Stirling interpolation formula:

- Ambil** (red arrow) points to $\Delta^2 f_{-1}$ in the second term.
- Rata-Rata** (red arrow) points to Δf in the first term.
- Ambil** (red arrow) points to $\Delta^3 f$ in the third term.
- Rata-Rata** (red arrow) points to $\Delta^5 f$ in the fifth term.
- Ambil** (red arrow) points to $\Delta^4 f_{-2}$ in the fourth term.

$$\begin{aligned}\Delta f &= \frac{1}{2} (\Delta f_{-1} + \Delta f_0) \\ \Delta^3 f &= \frac{1}{2} (\Delta^3 f_{-2} + \Delta^3 f_{-1}) \\ \Delta^5 f &= \frac{1}{2} (\Delta^5 f_{-3} + \Delta^5 f_{-2})\end{aligned}$$

$$s = \frac{x_s - x_0}{h}, \quad h = \Delta x$$

Tentang Interpolasi **Stirling**:

- Stirling merupakan rerata dari rumusan **Gauss Forward** dan **backward**. Karena menggunakan rerata beda gasal di atas dan di bawah **garis sentral**, serta beda genap pada garis sentral
- Kelebihan dan kekurangan metode Stirling sama dengan metode Newton-Gregory. Perbedaannya, Stirling lebih optimal jika digunakan untuk mencari nilai $f(x_s)$ dengan x_s di sekitar titik tengah

$\begin{matrix} > \\ = \\ < \end{matrix}$ **Tabel Beda** **Interpolasi Stirling** $\begin{matrix} < \\ = \\ > \end{matrix}$

f	Δf	$\Delta^2 f$	$\Delta^3 f$	$\Delta^4 f$	$\Delta^5 f$	$\Delta^6 f$	$\Delta^7 f$	$\Delta^8 f$
f_{-4}	Δf_{-4}							
f_{-3}		$\Delta^2 f_{-4}$						
	Δf_{-3}		$\Delta^3 f_{-4}$					
f_{-2}		$\Delta^2 f_{-3}$		$\Delta^4 f_{-4}$				
	Δf_{-2}		$\Delta^3 f_{-3}$		$\Delta^5 f_{-4}$			
f_{-1}		$\Delta^2 f_{-2}$		$\Delta^4 f_{-3}$		$\Delta^6 f_{-4}$		
	Δf_{-1}		$\Delta^3 f_{-2}$		$\Delta^5 f_{-3}$		$\Delta^7 f_{-4}$	
f_0		$\Delta^2 f_{-1}$		$\Delta^4 f_{-2}$		$\Delta^6 f_{-3}$		$\Delta^8 f_{-4}$
	Δf_0		$\Delta^3 f_{-1}$		$\Delta^5 f_{-2}$		$\Delta^7 f_{-3}$	
f_1		$\Delta^2 f_0$		$\Delta^4 f_{-1}$		$\Delta^6 f_{-2}$		
	Δf_1		$\Delta^3 f_0$		$\Delta^5 f_{-1}$		$\Delta^7 f_{-2}$	
f_2		$\Delta^2 f_1$		$\Delta^4 f_0$		$\Delta^6 f_{-1}$		
	Δf_2		$\Delta^3 f_1$		$\Delta^5 f_0$			
f_3		$\Delta^2 f_2$		$\Delta^4 f_1$				
	Δf_3		$\Delta^3 f_2$					
f_4		$\Delta^2 f_3$						
	Δf_4							
f_5								

↖ **Garis Sentral**

Interpolasi Bessel

Interpolasi Bessel:

$$\begin{aligned}
 f_n(x) = & f + u\Delta f_0 + \overset{\text{Rata-Rata}}{\frac{(u^2 - \frac{1}{4})}{2!}} \Delta^2 f + \overset{\text{Rata-Rata}}{\frac{u(u^2 - \frac{1}{4})}{3!}} \Delta^3 f_{-1} + \overset{\text{Ambil}}{\frac{(u^2 - \frac{1}{4})(u^2 - \frac{9}{4})}{4!}} \Delta^4 f + \overset{\text{Rata-Rata}}{\frac{u(u^2 - \frac{1}{4})(u^2 - \frac{9}{4})}{5!}} \Delta^5 f_{-2} + \dots
 \end{aligned}$$

$$s = \frac{x_s - x_0}{h}, \quad h = \Delta x$$

$$u = s - \frac{1}{2}$$

$$\begin{aligned}
 f &= \frac{1}{2} (f_0 + f_1) \\
 \Delta^2 f &= \frac{1}{2} (\Delta^2 f_{-1} + \Delta^2 f_0) \\
 \Delta^4 f &= \frac{1}{2} (\Delta^4 f_{-2} + \Delta^4 f_{-1})
 \end{aligned}$$

Tabel Beda Interpolasi Bessel

f	Δf	$\Delta^2 f$	$\Delta^3 f$	$\Delta^4 f$	$\Delta^5 f$	$\Delta^6 f$	$\Delta^7 f$	$\Delta^8 f$
f_{-4}								
	Δf_{-4}							
f_{-3}		$\Delta^2 f_{-4}$						
	Δf_{-3}		$\Delta^3 f_{-4}$					
f_{-2}		$\Delta^2 f_{-3}$		$\Delta^4 f_{-4}$				
	Δf_{-2}		$\Delta^3 f_{-3}$		$\Delta^5 f_{-4}$			
f_{-1}		$\Delta^2 f_{-2}$		$\Delta^4 f_{-3}$		$\Delta^6 f_{-4}$		
	Δf_{-1}		$\Delta^3 f_{-2}$		$\Delta^5 f_{-3}$		$\Delta^7 f_{-4}$	
f_0		$\Delta^2 f_{-1}$		$\Delta^4 f_{-2}$		$\Delta^6 f_{-3}$		$\Delta^8 f_{-4}$
	Δf_0		$\Delta^3 f_{-1}$		$\Delta^5 f_{-2}$		$\Delta^7 f_{-3}$	
f_1		$\Delta^2 f_0$		$\Delta^4 f_{-1}$		$\Delta^6 f_{-2}$		$\Delta^8 f_{-3}$
	Δf_1		$\Delta^3 f_0$		$\Delta^5 f_{-1}$		$\Delta^7 f_{-2}$	
f_2		$\Delta^2 f_1$		$\Delta^4 f_0$		$\Delta^6 f_{-1}$		
	Δf_2		$\Delta^3 f_1$		$\Delta^5 f_0$			
f_3		$\Delta^2 f_2$		$\Delta^4 f_1$				
	Δf_3		$\Delta^3 f_2$					
f_4		$\Delta^2 f_3$						
	Δf_4							
f_5								

Garis Sentral

Interpolasi Stirling

Contoh Soal 1

Diketahui Tabel sebagai berikut:

x	$f(x)$	Δf	$\Delta^2 f$	$\Delta^3 f$
3	-639,90			
		-680,95		
3,6	-1320,85		-411,63	
		-1092,58		-104,01
4,2	-2413,43		-515,64	
		-1608,22		-82,11
4,8	-4021,66		-597,76	
		-2205,98		-40,62
5,4	-6227,64		-638,38	
		-2844,36		23,83
6	-9072,00		-614,55	
		-3458,92		
6,6	-12530,92			

Carilah nilai $f(x)$, ketika $x = 4.9$ dengan menggunakan **Interpolasi Stirling Orde 3**

➤ Jawaban Contoh 1 ➤

• Stirling:

x	$f(x)$	Δf	$\Delta^2 f$	$\Delta^3 f$
3	-639,90			
		-680,95		
3,6	-1320,85		-411,63	
		-1092,58		-104,01
4,2	-2413,43		-515,64	
		-1608,22		-82,11
4,8	-4021,66		-597,76	
		-2205,98		-40,62
5,4	-6227,64		-638,38	
		-2844,36		23,83
6	-9072,00		-614,55	
		-3458,92		
6,6	-12530,92			

$$f_n(x) = f_0 + s\Delta f +$$

$$\frac{s^2}{2!}\Delta^2 f_{-1} + \frac{s(s^2 - 1)}{3!}\Delta^3 f$$

$$s = \frac{x_s - x_0}{h}, \quad h = \Delta x$$

$$s = \frac{4.9 - 4.8}{h}, \quad h = 4.8 - 4.2 = 0.6$$

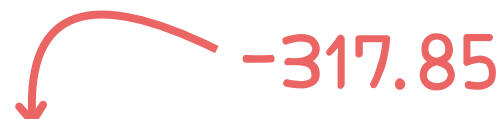
$$s = 0.17$$

$$\Delta f = \frac{-2205.98 - 1608.23}{2} = -1907.11$$

$$\Delta^3 f = \frac{-40.63 - 82.1}{2} = -61.36$$

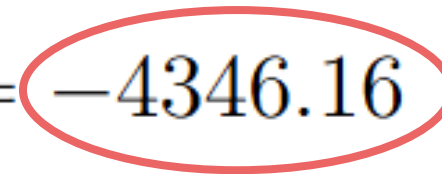
➤ Jawaban Contoh 1 ➤

$$f_n(x) = f_0 + s\Delta f + \frac{s^2}{2!}\Delta^2 f_{-1} + \frac{s(s^2 - 1)}{3!}\Delta^3 f$$

$$f_3(4.9) = -4021.66 + (0.17)(-1907.11) +$$


$$\frac{(0.17)^2}{2!}(-597.75) +$$


$$\frac{(0.17)((0.17)^2 - 1)}{3!}(-61.36)$$

$$= -4346.16$$


Interpolasi Bessel

Contoh Soal 1

Diketahui Tabel sebagai berikut:

x	$f(x)$	Δf	$\Delta^2 f$	$\Delta^3 f$
3	-639,90			
		-680,95		
3,6	-1320,85		-411,63	
		-1092,58		-104,01
4,2	-2413,43		-515,64	
		-1608,22		-82,11
4,8	-4021,66		-597,76	
		-2205,98		-40,62
5,4	-6227,64		-638,38	
		-2844,36		23,83
6	-9072,00		-614,55	
		-3458,92		
6,6	-12530,92			

Carilah nilai $f(x)$, ketika $x = 4.9$ dengan $x_0 = 4.8$ dan menggunakan **Interpolasi Bessel Orde 3**

➤ Jawaban Contoh 1 ➤

• Bessel:

x	$f(x)$	Δf	$\Delta^2 f$	$\Delta^3 f$
3	-639,90			
		-680,95		
3,6	-1320,85		-411,63	
		-1092,58		-104,01
4,2	-2413,43		-515,64	
		-1608,22		-82,11
4,8	-4021,66		-597,76	
		-2205,98		-40,62
5,4	-6227,64		-638,38	
		-2844,36		23,83
6	-9072,00		-614,55	
		-3458,92		
6,6	-12530,92			

$$f_n(x) = f + u\Delta f_0 +$$

$$\frac{(u^2 - \frac{1}{4})}{2!} \Delta^2 f + \frac{u(u^2 - \frac{1}{4})}{3!} \Delta^3 f_{-1}$$

$$s = \frac{x_s - x_0}{h}, \quad h = \Delta x$$

$$s = \frac{4.9 - 4.8}{h}, \quad h = 4.8 - 4.2 = 0.6$$

$$s = 0.17$$

$$u = s - \frac{1}{2} = -0.33$$

$$f = \frac{f_0 + f_1}{2} = -5124.65$$

$$\Delta^2 f = \frac{\Delta^2 f_{-1} + \Delta^2 f_0}{2} = -618.06$$

➤ Jawaban Contoh 1 ➤

$$f_n(x) = f + u\Delta f_0 +$$

$$\frac{(u^2 - \frac{1}{4})}{2!} \Delta^2 f + \frac{u(u^2 - \frac{1}{4})}{3!} \Delta^3 f_{-1}$$

$$f_3(4.9) = -5124.65 + (-0.33)(-2205.98) +$$

$$\frac{(-0.33)^2 - \frac{1}{4}}{2!} (-2205.98) +$$

$$\frac{(-0.33) \left((-0.33)^2 - \frac{1}{4} \right)}{3!} (-40.63)$$

$$= -4346.72$$

➤ Interpolasi Hermite ➤

$$\begin{aligned} f_n(x) = & \frac{\sin(x - x_1) \sin(x - x_2) \cdots \sin(x - x_n)}{\sin(x_0 - x_1) \sin(x_0 - x_2) \cdots \sin(x_0 - x_n)} + \\ & \frac{\sin(x - x_0) \sin(x - x_2) \cdots \sin(x - x_n)}{\sin(x_1 - x_0) \sin(x_1 - x_2) \cdots \sin(x_1 - x_n)} + \\ & \vdots \\ & \frac{\sin(x - x_0) \sin(x - x_1) \cdots \sin(x - x_{n-1})}{\sin(x_n - x_0) \sin(x_n - x_1) \cdots \sin(x_n - x_{n-1})} \end{aligned}$$

Tentang Interpolasi **Hermite**:

- Karena 'diturunkan' dari rumus Interpolasi Lagrange, maka kelebihan dan kekurangan dari interpolasi Hermite secara umum sama dengan Interpolasi Lagrange

Tugas Individu di kertas

1. Buatlah contoh soal sendiri, boleh mengarang atau mengambil dari internet:
 - a. Interpolasi newton – gregory forward orde 3
 - b. Interpolasi newton – gregory backward orde 3
 - c. Interpolasi stirling
 - d. Interpolasi Bessel



Komnum Week 8



TERIMA KASIH

Sampai Bertemu Kembali

